

A Fast Quantum Mechanical Algorithm for Database Search

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1. ABSTRACT & THEORETICAL FOUNDATION

We present a quantum search algorithm that locates a single marked item in an unsorted database of N items in $O(\sqrt{N})$ query iterations, achieving a provably optimal quadratic speedup.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Looking for a needle in a haystack of 1,000,000 items: classical search checks 500,000 items on average; Grover's algorithm finds it in just 1,000 checks by turning down the noise and amplifying the needle's signal."

3. Mathematical Formulation & Unitary Rule

$$G = (2|s\rangle\langle s| - I)(I - 2|\omega\rangle\langle\omega|), \quad R \approx \left\lfloor \frac{\pi}{2\theta} \right\rfloor$$

Geometric rotation on a 2D circle: each step adds 2θ toward the marked target state $|\omega\rangle$.

4. Step-by-Step Circuit Sequence

$H^{\otimes n}$ initialization, loop R times [Oracle $U_f \rightarrow H^{\otimes n} \rightarrow$ Phase Flip 0 $\rightarrow H^{\otimes n}$], measurement.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(2, 2)
# Step 1: Initialize equal superposition
qc.h([0, 1])

# Step 2: Oracle for target |11> (Controlled-Z)
qc.cz(0, 1)
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
$O(N)$ queries for unsorted list search	$O(\sqrt{N})$ queries (provably optimal)	Quadratic (black-box search)

7. Key Applications & Future Research Directions

- Database search and preimage attacks on cryptographic hash functions (SHA-256)
- Accelerating NP-complete constraint satisfaction problems (SAT)
- Subroutine in quantum collision finding and triangle finding

Future Work:

Application to combinatorial NP search problems, minimum finding, and amplitude amplification generalizations.